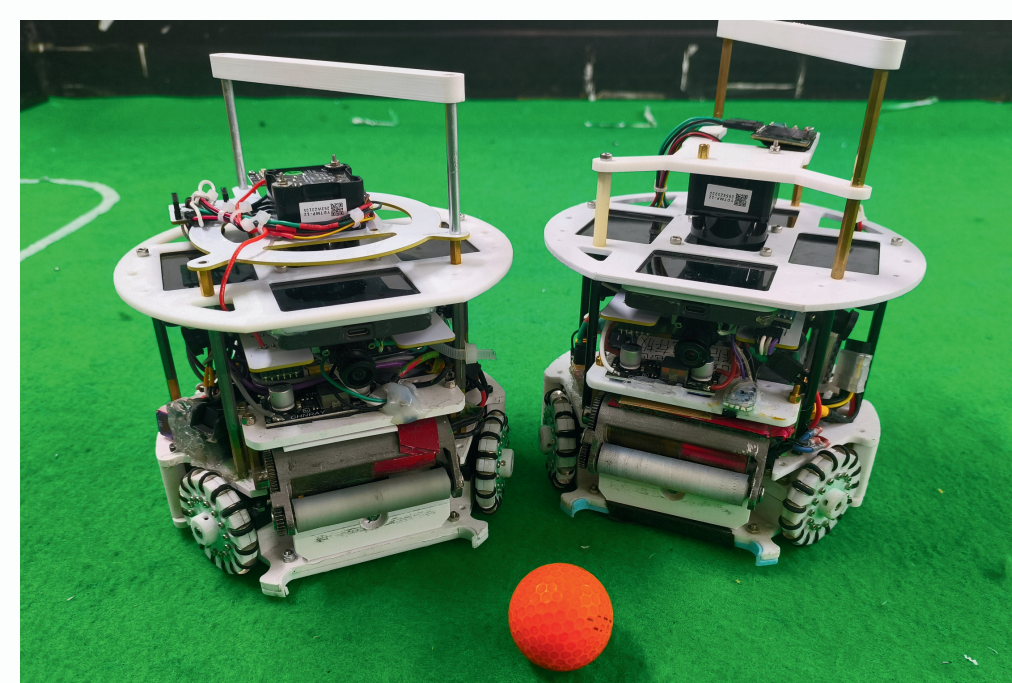


ABSTRACT

We are a Macau team participating in soccer Vison. Our team this year consists of three high school students. This year, we have made major changes to our bodies. We hope to play exciting games on the field, communicate with other players, and make progress together.

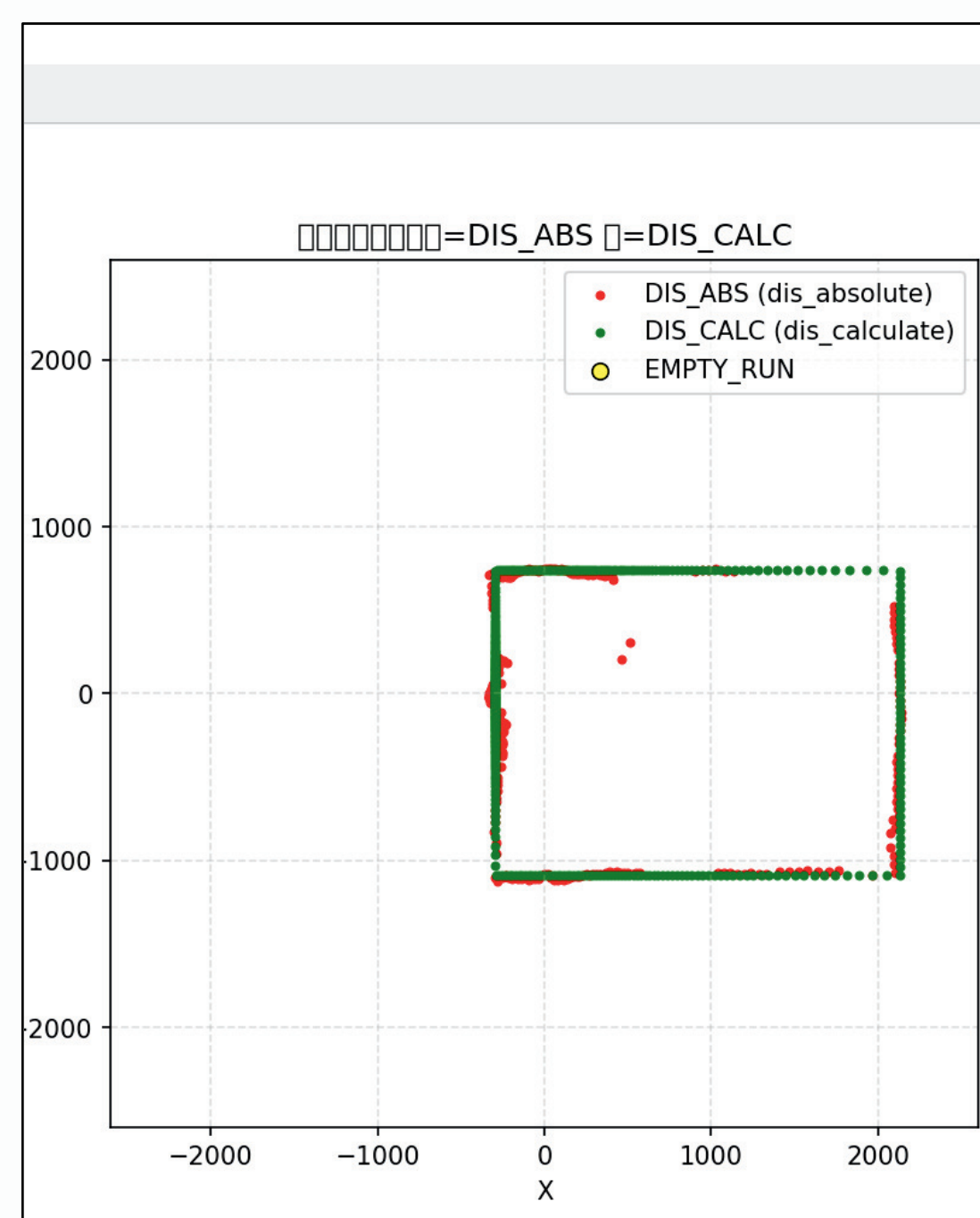


DESIGN / ALGORITHMS

Omnidirectional positioning system

-Use and development of lidar

We have achieved full-field localization based on LiDAR. The system collects environmental point cloud data via LiDAR and integrates gyroscope attitude data to correct each reflection point. All valid points are scored and voted on in both horizontal and vertical directions, and the four highest-scoring line segments are extracted to construct the complete field boundary.



Simultaneously, LiDAR can detect scattered interference points in the field and calculate the real-time coordinates of enemy robots, providing reliable positional information for strategic decision-making.

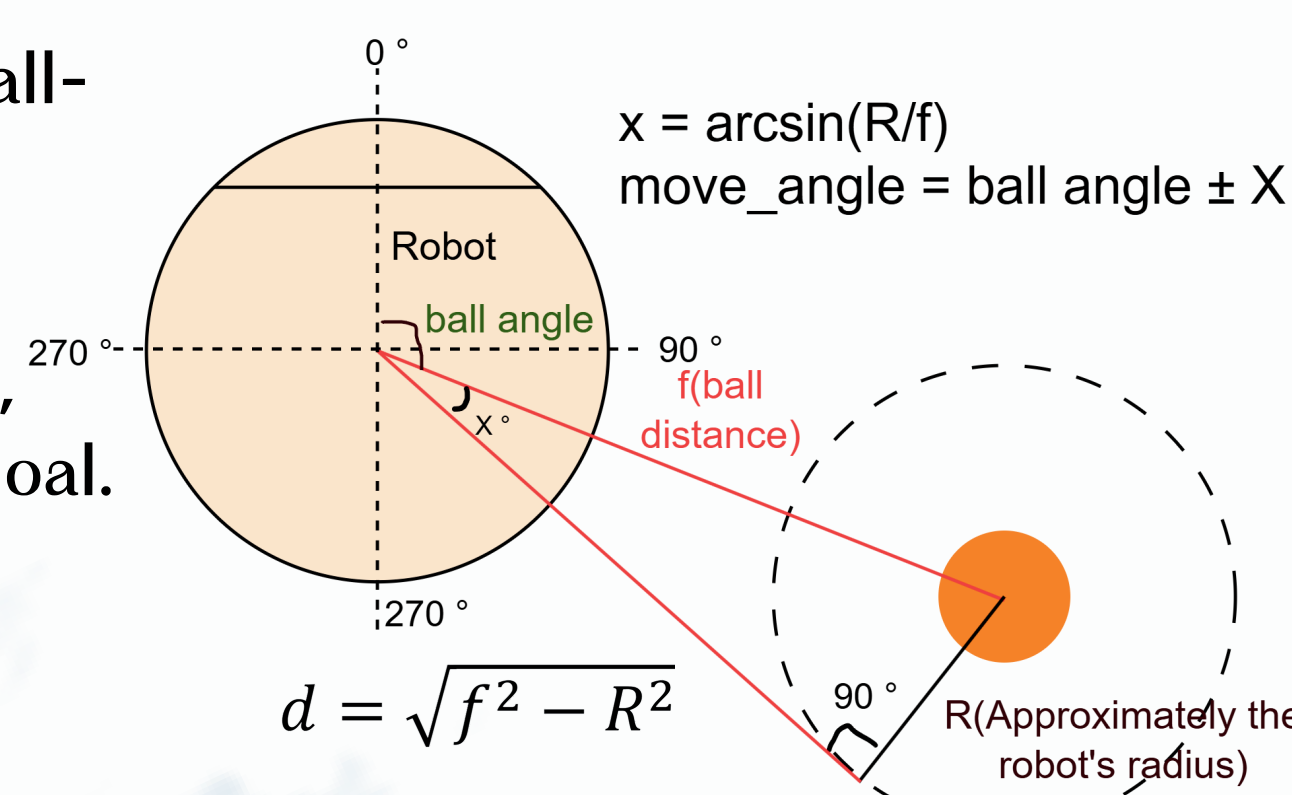
Our two robots communicate via Bluetooth, sharing their own status. By filtering out point cloud interference generated by the teammate robot, stable and accurate enemy coordinate localization is achieved.

Long-distance vision system

Our robot is equipped with four cameras to deliver all-round vision, expanding ball detection range and eliminating visual blind spots.

We use tangent circular trajectories to orbit the ball, enabling the shortest moving path while facing the goal. We set a virtual circle around the ball and compute distances and steering angles with right triangle geometry and inverse trigonometric functions. Calculation details are shown in the attached figure.

-Use of 4 cameras

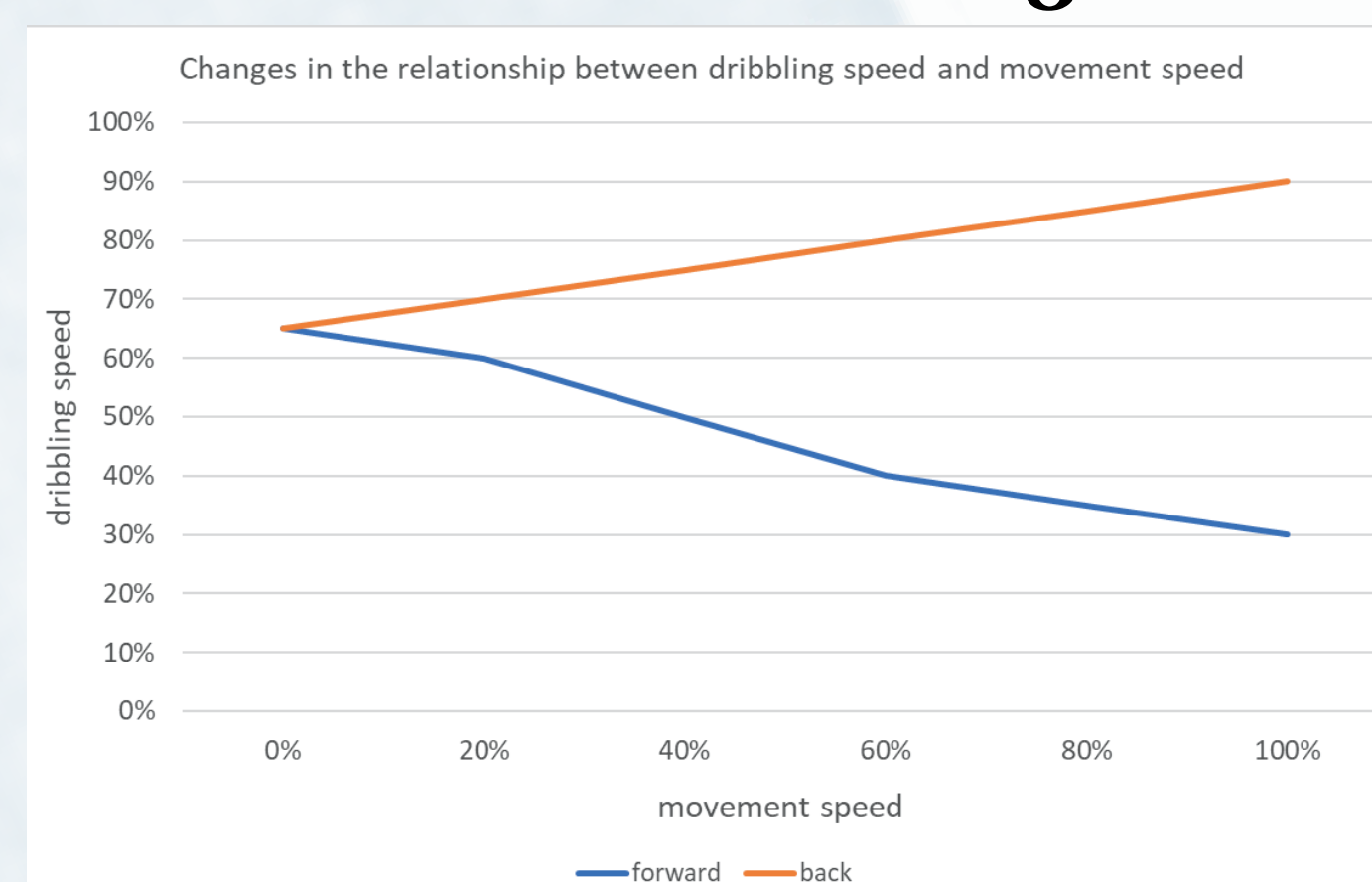


Speed control system

The robot's field movement speed is regulated via PID control for both linear and angular velocity. When the robot moves forward without shortening its distance to the ball, the controller automatically increases speed to approach faster.

Meanwhile, variable dribble speed is implemented: the dribbler's speed adjusts dynamically with the robot's angular velocity to prevent ball loss during rotation.

-Use of PID Regulation



Relationship Diagram of Dribbler Speed and Robot Motion Speed

OVERALL

Positioning module

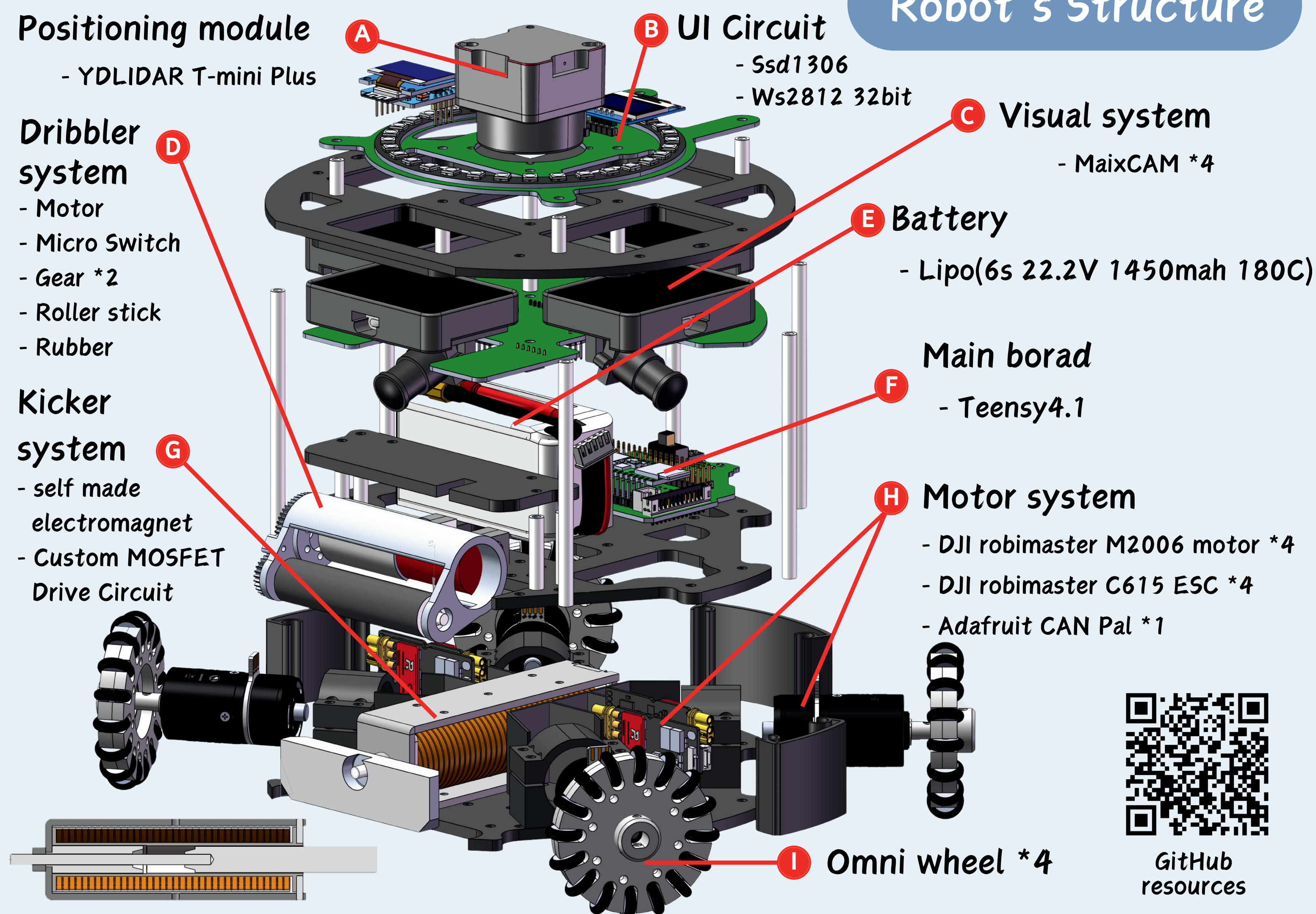
- YDLIDAR T-mini Plus

Dribbler system

- Motor
- Micro Switch
- Gear *2
- Roller stick
- Rubber

Kicker system

- self made electromagnet
- Custom MOSFET Drive Circuit

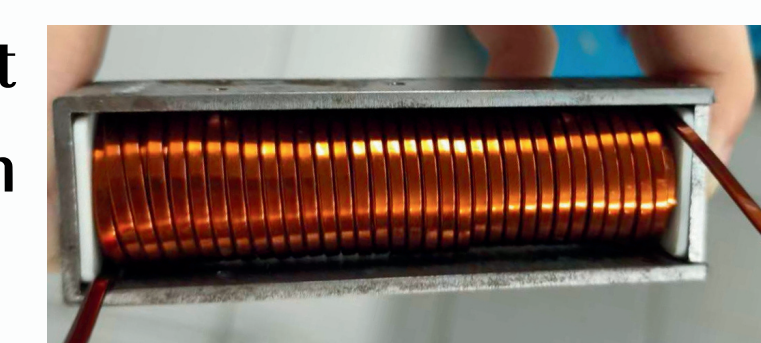


EXPERIMENT / DATA

Kicker system

- Experiments & Data of Custom Electromagnet

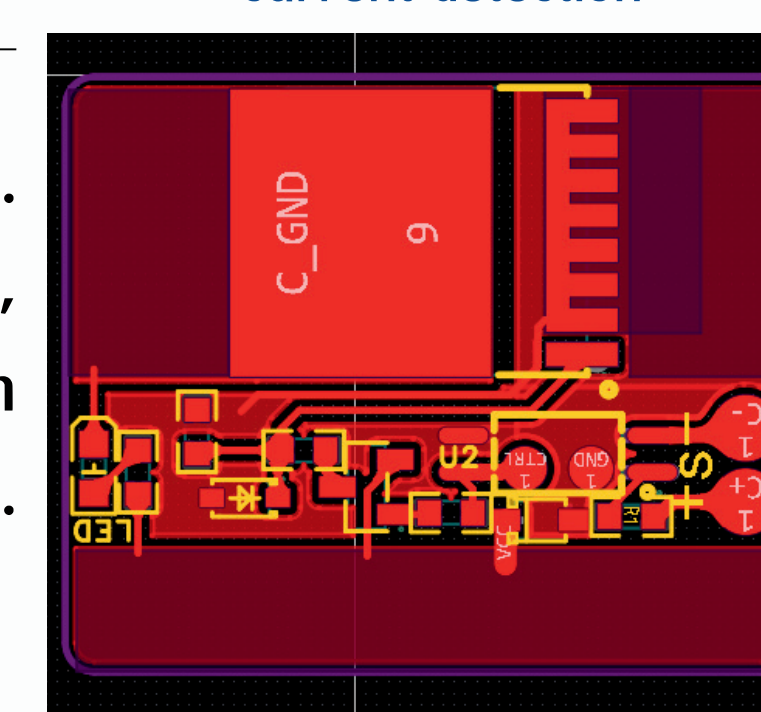
We conducted in-depth research on the coil and even built one ourselves. We found that increasing the current gain seemed to be greater than increasing the number of turns.



Through derivation and testing, we determined the complete manufacturing process. Flat enameled wire was used for the winding to improve space utilization, and the coil resistance was verified using a milliohm meter after winding.



We developed a dedicated MOS drive circuit for this high-power coil. High-current MOSFETs have large parasitic capacitances; we added a push-pull circuit to accelerate the gate charging and discharging speed, avoiding MOSFET thermal breakdown due to switching losses during ramp-up. An external Schottky diode was used to prevent backflip, and VGS and current were monitored in real-time using an oscilloscope. A fuse was also installed for fault protection. Drawings and circuit diagrams are available on GitHub.



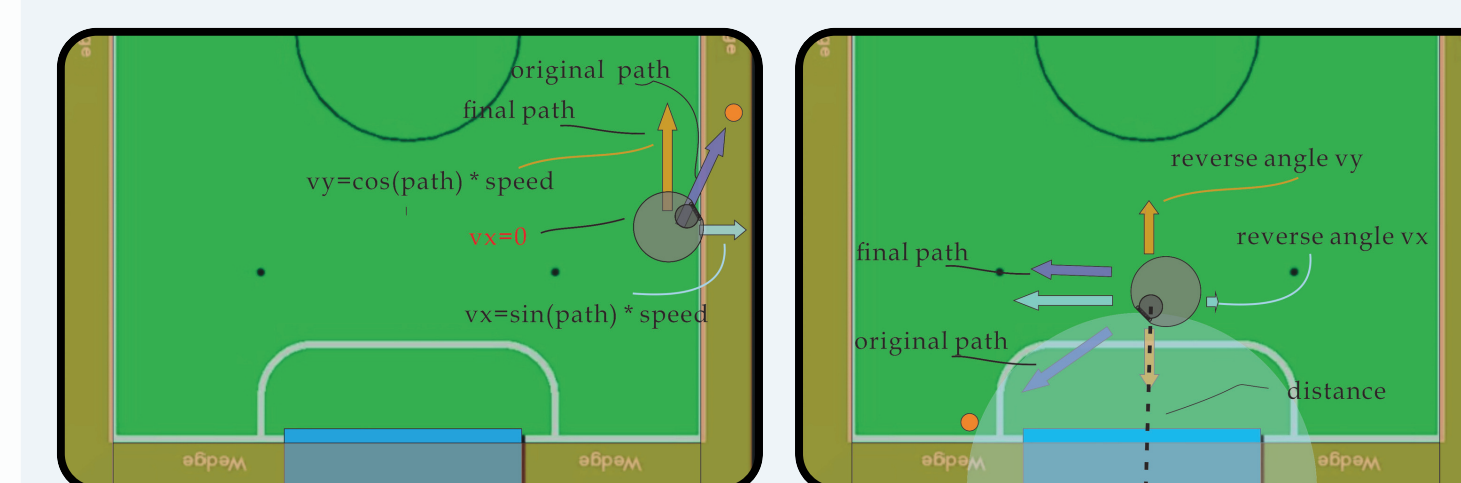
To standardize the experiments, we built a portable velocity measurement device. The flight time of a golf ball between two fixed laser beams was recorded, and its launch velocity was calculated to quantify the electromagnetic output performance. Currently, our electromagnet can achieve a velocity of approximately 3.8 m/s on this device.



STRATEGY

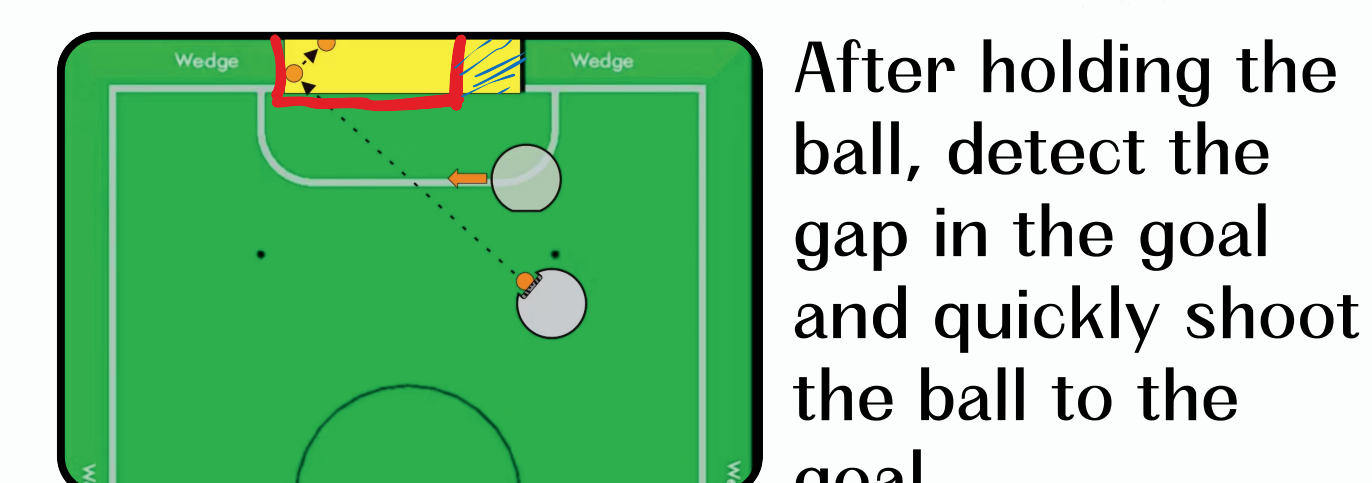
(The video can be found on our YouTube channel.)

Out of bounds



When the robot touches our restricted area (sideline and restricted area), we decompose the speed into the x-axis and y-axis components, then cancel the out-of-bounds direction and re-synthesize to get the final speed.

Offense

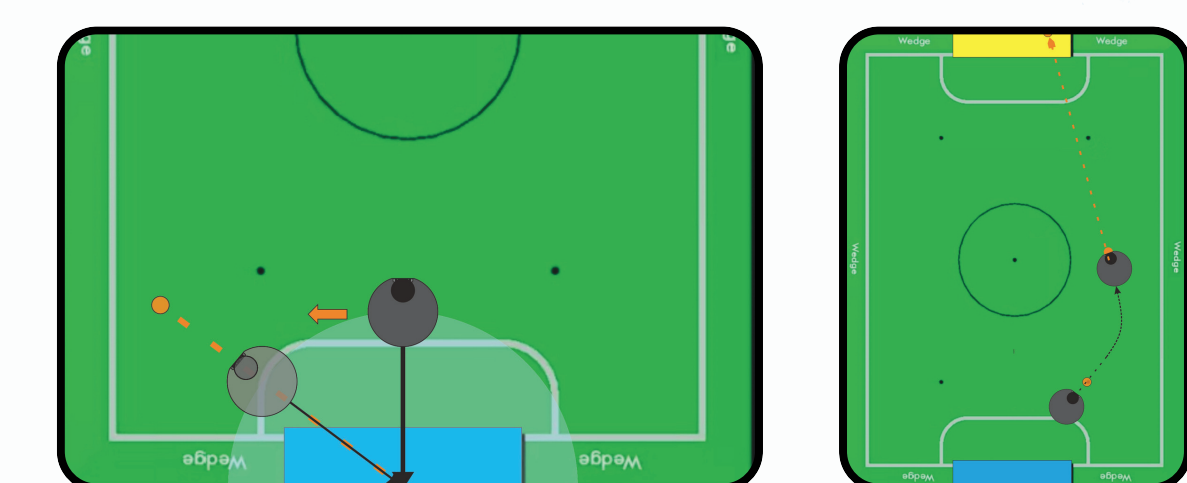


After holding the ball, detect the gap in the goal and quickly shoot the ball to the goal.



then turn around and shoot the goal.

Deffense



Establish a circle as a defensive movement trajectory, and stay on the line connecting the ball and the center of the goal, limiting the opponent's maximum offensive space, with the head always facing outward. When the ball is stationary, it will dribble and then shoot the ball towards the goal.

Environment

